

July 2025, CSE 106

Online on Arrays

Section: A1+A2

Time: 40 minutes

You are given an array **A** of monthly profit/loss values for a company. For any pair of month indices $i \leq j$, define $\text{sum}[i, j] = A[i] + A[i+1] + \dots + A[j]$. Your goal is to find a time interval $[i, j]$ that maximizes this total profit.

Your algorithm must run in **$O(n)$** time and **cannot create any new arrays**.

If multiple intervals have the same maximum total profit, you may return any one of them.

Return the start index i and end index j .

You are free to tackle the problem however you like, but that approach will earn at most 60%. To receive full marks, you need to meet the stated time and space constraints.

Example 1:

Input: $A = [4, -3, 5, -7, 9, -2, 5, -7, 9, -2, -5, 6]$

Output: $[4, 8]$

Explanation: $A[4] + A[5] + \dots + A[8] = 16$, which is the window with maximum sum.

Example 2:

Input: $A = [-2, -1, -3, -4]$

Output: $[1, 1]$

Explanation: This is the only possible window of length 1, including any more elements will only decrease the sum.

Example 3:

Input: $A = [5, 4, -10, 3, 2]$

Output: $[0, 1]$

Example 4:

Input: $A = [1, 2, 3, 4]$

Output: $[0, 3]$

Example 5:

Input: $A = [8, -1, 2, -1, 10, -20, 5]$

Output: $[0, 4]$